

PRES1 Speaking Guide (v2)

PathCompanion AI — Slide-by-Slide: Enna Pesanum (Tamil + Simple English)

Deck theme: EVLO WORK + ENNA SPECIAL. Target 10–15 min. Marking: Demo 45% ·
Knowledge 20% · Contribution 15% · Skills 15% · Flow 5%

Slide 1: Title (~30 sec)

SAY THIS (simple English)

"Good morning. I am [name]. My project is PathCompanion AI — an agentic career companion. In one line: six AI agents, two machine-learning models that I trained myself, one application, zero running cost. Over the next fifteen minutes I will show you what I built, why it is different, and then demonstrate it live."

TAMIL-LA (nee purinjukka)

Confident opening: un peru, project peru, antha punch line — '6 agents, naan train pannina 2 models, 1 app, 0 cost'. 'Enna build pannen, ethuku ithu different, live-ah kaattren' nu roadmap kudu.

TIP: First 30 seconds set the tone — memorise this opening completely.

Slide 2: Background — Why This Project (~1 min)

SAY THIS (simple English)

"Look at what a job seeker does today: searches on three or four job boards, guesses which skills they lack, learns from random courses, sends one resume everywhere — which ATS filters reject — and walks into interviews anxious, with no practice. Five disconnected steps, and nothing done in one step helps the next. Career coaching could fix this, but it costs more than most graduates can afford. So my idea was: one intelligent application where every part talks to every other part — for free."

TAMIL-LA (nee purinjukka)

Anta 5 boxes-ah point pannitte oru job seeker-oda kastam-ah story-ah sollu. Punch: 'onnu-la pannrathu innonnu-ku help aagala'. Coaching cost-ah thaanga mudiyala — athaan free-ah oru integrated app build pannen.

Slide 3: Problem Statement (~45 sec)

SAY THIS (simple English)

"I broke that pain into six precise problems: inefficient job discovery, invisible skill gaps, unguided learning, generic resumes, unaddressed interview anxiety, and opaque career progression. The design answer is one agent per problem — and all six connected through shared memory."

TAMIL-LA (nee purinjukka)

6 problems-um fast-ah sollu — details screen-la irukku, read panna vendaam. Mudivu-la: 'ovvoru problem-ku oru agent, ella agents-um shared memory-la connect' — ithu next slides-ku bridge.

Slide 4: What I Built — Six Agents, One Brain (~1.5 min)

SAY THIS (simple English)

"Here is the system. Job Scout pulls real jobs from public APIs and matches them semantically to your profile. Skill Gap Analyzer — powered by a model I trained — reads any job ad and shows exactly what you have and what you miss. Learning Finder turns every gap into an ordered plan of free resources. Resume Architect classifies your resume with my other trained model and generates an ATS-optimised version per job. The Voice Mock Interview actually speaks to you, listens, and coaches — with a panic button for anxiety. And Career Path Planner maps your road from intern to senior. On top of the proposal, I also delivered a mock exam module with certificates and a cross-app

assistant."

TAMIL-LA (nee purinjukka)

Ithu un product showcase slide — ovvoru agent-aiyum oru vaakiyam-la sollu, aana 'model naan train pannaen' nu rendu edathula azhutham kudu. Last line important: 'proposal-ku mela extra-va mock exam + assistant deliver pannirukken'.

Slide 5: The Work Behind It (~1 min — the EFFORT slide)

SAY THIS (simple English)

"Let me show you the scale of the work. Sixteen weeks, eight sprints, each ending with something working. Two machine-learning models trained from scratch — not API calls, actual training on over fourteen thousand labelled samples. Thirteen database tables, all secured with row-level security. Nine backend routers with eleven services. Eleven frontend pages. Five different AI services orchestrated together. And the total running cost: zero. Every piece — design, code, training, testing, documentation — done by one person."

TAMIL-LA (nee purinjukka)

Ithu 'evlo work' slide — numbers-ah proud-ah sollu, avasarpadaathe. '14,300 labelled samples', '13 tables', '0 cost' — intha moonu numbers-ah kandippa sollanum. Last line: 'ellame oruthar than pannirukken' — confident-ah, aana arrogant illa.

TIP: Pause after '0 rupees' — let it land. This slide earns respect for effort.

Slide 6: Special #1 — The Cascade (~1.5 min)

SAY THIS (simple English)

"Now the three things that make this project special. First: my agents react to each other. Watch the flow — you mark Docker as learned, an event fires in shared memory, the Skill Gap agent re-checks itself using my trained model, and your resume gets flagged for refresh. One click by the user; everything else is automatic, under thirty seconds, and the screen never freezes. Technically, agents never call each other — they communicate through an event bus, which means every module also works standalone, and the whole thing is crash-safe. You will see this live in my demo."

TAMIL-LA (nee purinjukka)

USP #1. Story format: 'Docker learn pannitten nu oru click → background-la ellam thaanaa nadakkuthu'. Technical line oru thadava mattum: 'agents direct-ah call pannaathu, event bus vazhiya' — viva-la ithu pathi kekkum bothu expand pannalaam. 'Demo-la live-ah kaattren' nu promise pannu.

Slide 7: Special #2 — My Own Trained Models (~1.5 min)

SAY THIS (simple English)

"Second special thing — and this is the academic heart: I did not just call an AI API. I trained my own models. Model 1 reads any resume and identifies its career field out of twenty-four categories — trained on nearly two and a half thousand labelled resumes, baseline first, then a fine-tuned transformer, reaching 82.3 percent accuracy, beating my own baseline by almost seventeen points. Model 2 detects which skills a job ad demands. No labelled data existed for this — so I generated the labels myself for nearly twelve thousand postings using LLM-assisted weak supervision, a technique from current research literature. The result: the core intelligence of my app is mine — reproducible,

benchmarked, and independent of any external provider."

TAMIL-LA (nee purinjukka)

USP #2 — academic marks inga thaana. Structure: 'API kooppudala, naan train pannen'. Model 1: numbers sollu (82.3%, +16.8 points). Model 2: 'labels illa, naan-e generate pannen — research technique' nu highlight pannu. Mudivu: 'core intelligence ennodathu' — strong line, confidence-oda sollu.

Slide 8: Special #3 — It Speaks, Listens, and Cares (~1 min)

SAY THIS (simple English)

"Third: the human side. The interview agent speaks questions aloud — tailored to the real job and your real skills — listens to your spoken answer, and coaches you, with about two seconds per turn, so it feels like a real conversation. The panic button is my favourite feature: interview anxiety is real, so one press pauses everything and guides breathing exercises — and the app measures anxiety before and after each session. And all of this — speech recognition, neural voices, two LLMs, GPU training — runs entirely on free tiers. A student in Sri Lanka pays nothing."

TAMIL-LA (nee purinjukka)

USP #3 — emotional angle. 'App pesuthu, kekkuthu, kandukkuthu (cares)'. Panic button-ah personal-ah sollu — 'interview anxiety real problem, athukkaaga naan design pannen'. Zero cost point-ah Sri Lanka context-oda mudi — examiner-ku social value theriyum.

Slide 9: Requirements & Methods (~1 min)

SAY THIS (simple English)

"Briefly, how this was organised. Requirements were prioritised with MoSCoW — the Musts cover the full core journey you just saw; panic control and the assistant were Shoulds; the mock exam was only a Could, but I delivered it anyway. Non-functional requirements: database-level security, two-second interview turns, a UI that never blocks, and one hundred percent free-tier operation. The method was Agile Scrum — eight two-week sprints, with two full sprints dedicated to machine learning, and every sprint ending with something demonstrable for my supervisor."

TAMIL-LA (nee purinjukka)

Compliance slide — brief kekkura 'requirements + development methods' inga cover aagum. Fast-ah po, aana 'Could-ah irundha mock exam-aiyum deliver pannitten' line-ah miss pannaathe. 'Two sprints full-ah ML-ku' — effort point.

Slide 10: Findings — Did It Work? (~1.5 min)

SAY THIS (simple English)

"So, did it work? The chart shows Model 1 on unseen test data: my fine-tuned transformer reached 82.3 percent accuracy against the 65.5 percent baseline — a 16.8-point improvement, which is the core scientific claim. Macro-F1 came to 0.751 against my 0.80 target — and I know exactly why: long resumes get truncated at 256 tokens, and rare categories drag the macro average; the fixes are longer inputs and class weighting. On the system side: the cascade verified under thirty seconds, interview turns around two seconds, everything on free tiers, and all twelve planned objectives achieved plus one module beyond scope. Every number here comes from a versioned, reproducible notebook."

TAMIL-LA (nee purinjukka)

Results-ah confident-ah sollu: 82.3 vs 65.5. Target miss-ah NEE-YA sollu, reason + fix ready-ah irukku — examiners-ku ithu critical evaluation skill-ah theriyum, marks kudupaanga. '12 objectives + 1 extra' nu positive-ah mudi.

TIP: Presenting your own miss with reasons = maturity marks. Never wait for them to find it.

Slide 11: Live Demonstration (~5-6 min — THE 45%)

SAY THIS (simple English)

"Now let me show you the real thing, in eight steps. One: I log in — the full stack is live. Two: my profile and skills — the shared memory. Three: Job Scout — real jobs with semantic match percentages. Four: Skill Gap — my Model 2 working live on a real job ad. Five — the highlighted step: I mark a skill as learned, and you watch the cascade ripple — the gap analysis refreshes and the resume gets flagged, automatically. Six: Resume classification — my Model 1 live with confidence scores. Seven: a voice interview turn — it speaks, I answer, and the panic button. Eight: the bonus mock exam with its certificate."

TAMIL-LA (nee purinjukka)

45% marks inga. App munnadiye run pannu, data ready-ah vachchuko. Step 5 (cascade)-ah slow-ah, drama-voda kaattu — 'naan ippo onnume click pannala paarunga' nu sollitu refresh aagurathai kaattu. Mic-ah munnadiye test pannu. Onnu fail aanaa panic aagaathe — 'normally ippadi nadakkum' nu solli backup screenshot kaattu.

TIP: Rehearse twice end-to-end with a timer. Keep 5–6 minutes strictly for this.

Slide 12: Contribution to Knowledge (~1 min)

SAY THIS (simple English)

"What does this work contribute? Engineering: a working blueprint for multi-agent AI over shared memory and events — reusable in healthcare, education, anywhere. Machine learning: a reproducible hybrid pattern — train your own models for frequent, well-defined tasks; keep LLMs for open conversation; benchmark everything. Methodology: proof that a single student can apply LLM-assisted weak supervision end-to-end, from raw public data to a served model. And social impact: career coaching for people who could never afford it, with anxiety-aware interview practice, at zero cost by design."

TAMIL-LA (nee purinjukka)

15% direct marks. Naalu contribution-um oru breath-la: engineering blueprint, ML hybrid pattern, weak supervision methodology, social impact. Ovvonnukkum oru vaakiyam pothum, aana naalayum sollanum.

Slide 13: Improvements & Future (~45 sec)

SAY THIS (simple English)

"What would I improve? Model 1 needs longer inputs and class weighting to hit its F1 target. Model 2 should grow from 25 skills to the wider ESCO vocabulary. Weak labels carry some noise, controlled by a gold-standard subset. Looking forward: a daily journal agent, one-click apply with the tailored resume, a recruiter portal, Tamil and Sinhala interfaces, and a mobile app on the same backend. This

is not a semester throwaway — it is a releasable product with a real path to real users."

TAMIL-LA (nee purinjukka)

Improvements-ah honest-ah, future-ah ambitious-ah sollu. Tamil/Sinhala interface sollumbothu adhu un own community-kkaana feature nu theriyum. Last line strong-ah: 'ithu semester throwaway illa — release panna mudiya product'.

Slide 14: Thank You / Q&A; (~Q&A;)

SAY THIS (simple English)

"Thank you very much. I am happy to take your questions."

TAMIL-LA (nee purinjukka)

Nimirndhu nillu, smile pannu, kelvigalai welcome pannu. Keezha irukkura Q&A; list-ah munnadiye practise pannirundhaa bayam varaathu.

Likely VIVA Questions — Ready Answers

Q: Why did you train your own models instead of just using an LLM?

EN: Three reasons: deterministic, reproducible results for the most frequent tasks; no external dependency or rate limits for core features; and a quantitatively evaluable contribution — I can benchmark my model against a baseline; an API call cannot be evaluated that way.

TA: LLM mattum use pannaa deterministic illa, rate limits, evaluate pannavum mudiyaathu. Own model train pannaa baseline-oda scientific compare panna mudiyum — athu thaana academic contribution.

Q: Why DistilBERT and not a bigger model?

EN: DistilBERT keeps ~97% of BERT's capability with 40% fewer parameters — it trains on a free Colab GPU and serves from a free-tier backend. Bigger models would break both my cost and deployment constraints.

TA: Free Colab-la train aaganum, free backend-la serve aaganum — DistilBERT thaana sweet spot. 97% capability, 40% kammi size.

Q: Why is macro-F1 below your target?

EN: Long resumes get truncated at 256 tokens, and the dataset is imbalanced — rare classes like BPO drag the macro average, which weights all classes equally. Fixes: 512-token inputs, class weighting, more epochs. The headline claim — +16.8 points over baseline — is unaffected.

TA: 256-token truncation + class imbalance. Fix ready: 512 tokens, class weighting. Main claim (+16.8 points) affect aagala.

Q: How does the cascade avoid blocking the UI?

EN: The API call does two fast writes — update the skill, insert an event — and returns immediately. Background consumers poll the event table's processed flag. Since consumers read from a table rather than being invoked, the cascade is idempotent and resumes safely after a crash.

TA: API rendu fast writes mattum pannitu udane return. Background-la event polling. Table-based-ah irukkurathala crash aanaalum resume aagum — idempotent.

Q: How is user data protected?

EN: Row-level security on every per-user table — `auth.uid() = user_id`, enforced by PostgreSQL itself. The job catalogue is shared read-only. API keys stay server-side. And no user data ever trains the models — only public research datasets.

TA: Database-level RLS — PostgreSQL-ye enforce pannuthu. User data model training-ku pogave pogaathu — public datasets mattum.

Q: What happens if free-tier APIs fail?

EN: Core analysis runs on my locally served models, so it keeps working. LLM features degrade gracefully — cached results, background retries. That resilience is exactly why the hybrid design puts my trained models on the most frequent path.

TA: Core features local models-la odum — continue aagum. LLM features cache + retry. Athukkaaga thaana hybrid design.

Q: How would this scale?

EN: Stateless FastAPI scales horizontally; HNSW vector indexes keep search fast; the event-driven cascade already decouples heavy work. At scale I would swap polling for a proper queue and upgrade tiers — the architecture itself does not change.

TA: Backend stateless — horizontal scale. Event bus already decoupled. Queue + paid tier maathanum, architecture apdiye irukkum.

Q: *What was the hardest part?*

EN: Model 2 — because no labelled data existed. Designing the weak-supervision pipeline, validating label quality, and making multi-label training converge taught me more than any other part of the project.

TA: Model 2 than kashtam — labelled data illave illa. Weak supervision pipeline design pannrathu, label quality check pannrathu — athula than naan romba kaththukitten.

Q: *What did you learn?*

EN: Baseline-first ML — build the simple model before the transformer or you cannot prove value. Set targets after exploring the data, not before. And event-driven decoupling repays its design cost many times over in testing safety.

TA: Baseline first. Targets EDA-kku appuram. Event-driven design testing-ah safe aakkuthu. Intha moonu than main lessons.